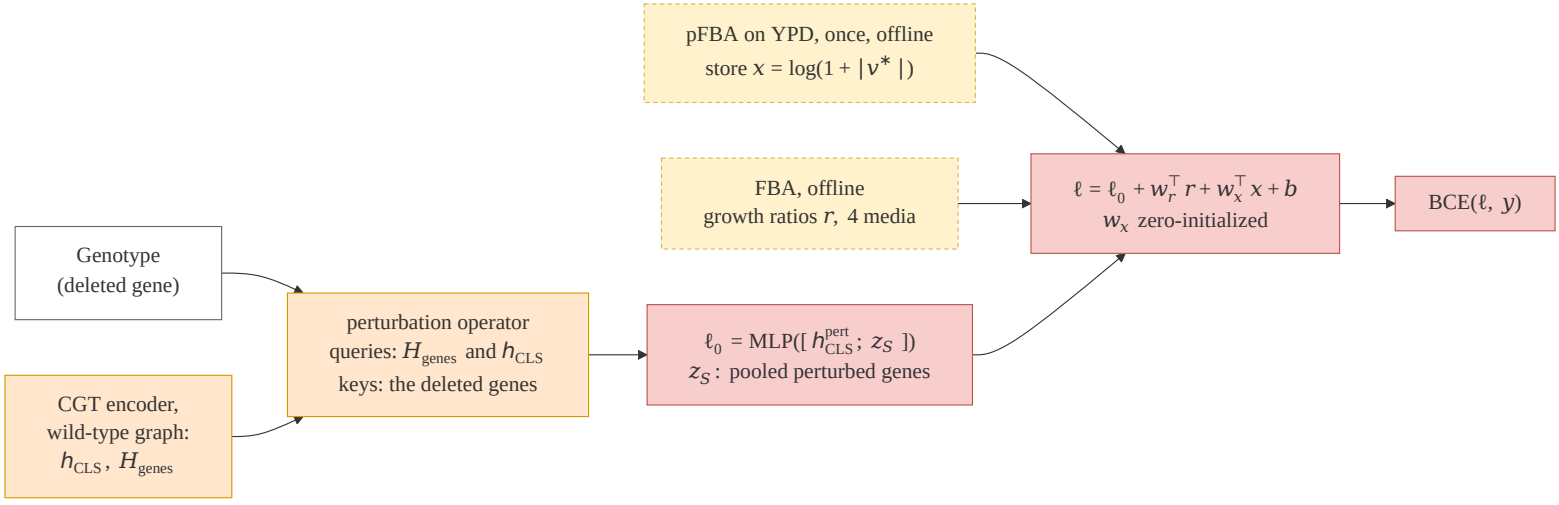

pFBA on YPD, once, offline
store $x = \log(1 + |v^*|)$
FBA, offline
growth ratios r, 4 media
$\ell = \ell_0 + w_r^\top r + w_x^\top x + b$
w_x zero-initialized
BCE(ℓ, y)
Genotype
(deleted gene)
perturbation operator
queries: H_{genes} and h_{CLS}
keys: the deleted genes
ℓ_0 = MLP([$h_{\text{CLS}}^{\text{pert}}$; z_S])
z_S : pooled perturbed genes
CGT encoder,
wild-type graph:
h_{CLS} , H_{genes}